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REVISIONS						
SYM	ZONE	DESCRIPTION	DR	CHK	DATE	APPROVED
		INITIAL RELEASE CLASS A PER TDRR 36704			8/20/68	S4B

REQUIREMENTS:

1. GENERAL:
   - A. INTERPRET DRAWING IN ACCORDANCE WITH THE STANDARDS PRESCRIBED BY MIL-D-70327.
   - B. SUPPLIER SHALL CONFORM TO THE QUALITY ASSURANCE PROVISIONS CONTAINED IN ND 1015404, CLASS 2.
   - C. UNITS SHALL BE CAPABLE OF MEETING ALL QUALIFICATION REQUIREMENTS OF ND 1002056.
   - D. PART MARKING: PARTS SHALL BE PERMANENTLY AND LEGIBLY MARKED IN ACCORDANCE WITH ND 1002019 WITH THE MANUFACTURER'S NAME AND/OR SYMBOL, LOT CODE OR NUMBER TERMINAL IDENTIFICATION AND NASA PART NUMBER (DRAWING NUMBER) AND REVISION LETTER.
   - E. PREPARATION FOR DELIVERY SHALL BE IN ACCORDANCE WITH ND 1002215, CLASS I, CODE 1.
     - (1) MARKING OF SHIPPING CONTAINERS SHALL CONFORM TO THE MARKING OF UNIT AND INTERMEDIATE PACKAGES AND THE METHODS OF MARKING AS SPECIFIED IN ND 1002215.

2. ACCEPTANCE AND INSPECTION: 100% EXCEPT AS OTHERWISE NOTED.
   - A. MECHANICAL AND PHYSICAL PROPERTIES:
     - (1) DIMENSIONS: AS DELINEATED HEREIN.
     - (2) SURFACE ROUGHNESS: 125√ UNLESS OTHERWISE SPECIFIED.
     - (3) MARKING: AS SPECIFIED IN NOTES 1.D AND 1.E. NO MARKING SHALL APPEAR ON THE 32√ AREA.
     - (4) PANEL COLOR AND MARKING: SAMPLE

       COLOR, ENERGIZED: CROSS HATCHED LEGEND AREAS TO BE YELLOW. ALL OTHER LEGEND AREAS TO BE WHITE.

       MARKING; LEGEND: LEGENDS SHALL BE AS SHOWN ON DRAWING. CHARACTERS SHALL BE GORTON CONDENSED WITH PROPORTIONS PER ND 1002122, TYPE II, CLASS 2. THE OVERALL HEIGHT OF CHARACTERS SHALL BE .156. THE LINE STROKE WIDTH SHALL BE .025/.030. LEGEND SHALL BE CENTERED ON DISPLAY FACE PLATE. FILL SHALL BE BLACK COLOR NO. 37038 PER FED-STD-595 AND SHALL BE OPAQUE WHEN LEGEND IS ENERGIZED.
     - (5) BRIGHTNESS: LEGEND BACKGROUNDS SHALL HAVE A BRIGHTNESS LEVEL OF 15 ± 3 FOOT-LAMBERTS AT 5.0 ± 0.05 VDC. THE LEGENDS MUST BE READABLE (WHEN ILLUMINATED) UNDER AN AMBIENT 50 FOOT CANDLE ILLUMINATION.
     - (6) ISOLATION: THERE SHALL BE NO LIGHT LEAKAGE FROM LIT TO UNLIT LEGEND AREAS, PAINT SHALL NOT BE DEEMED AN ADEQUATE LIGHT BLOCK.
     - (7)* MAXIMUM HOLE DIAMETER DIMENSIONS INDICATED ARE APPLICABLE FROM SURFACE TO .190 DEPTH ONLY. (SEE DETAIL A).
   - B. ELECTRICAL CHARACTERISTICS:
     - (1) POWER DISSIPATION:
       YELLOW DISPLAYS: 0.180 AMPERES ± 15% AT 5.0 ±.0.1 VDC.
       WHITE DISPLAYS: 0.225 AMPERES ±15% AT 5.0 ± 0.1 VDC.
     - (2) DIELECTRIC WITHSTANDING VOLTAGE: 200 ± 25 VRMS AT 60 CPS FOR 5 SECONDS APPLIED TO THE TERMINALS OF AN INDIVIDUAL LAMP CIRCUIT SHORTED TOGETHER AND ANY OTHER SIMILARLY SHORTED LAMP CIRCUIT TERMINAL AND BETWEEN ALL TERMINALS (EXCEPT 15) SHORTED TOGETHER AND THE CASE.
     - (3) INSULATION RESISTANCE: 100 MEGOHMS MINIMUM AT 100 VDC AT SEA LEVEL, 70° ± 10° F, 50% RELATIVE HUMIDITY MAXIMUM, BETWEEN THE TERMINALS OF ANY INDIVIDUAL LAMP CIRCUIT SHORTED TOGETHER AND ANY OTHER SIMILARLY SHORTED LAMP CIRCUIT TERMINAL AND BETWEEN ALL TERMINALS (EXCEPT 15) SHORTED TOGETHER AND THE CASE.

NOTE APPLICATION	
DASH NO.	APPLICABLE NOTES
-001	PARAGRAPHS 1 THROUGH 7

PROCURE ONLY FROM APPROVED SOURCES LISTED IN ND 1002034 FOR THIS DRAWING.

QTY REQD	PART OR IDENTIFYING NO.	MATERIAL OR NOTES	NOMENCLATURE OR DESCRIPTION	FIND NO.

LIST OF MATERIALS

NEXT ASSY	USED ON
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CAPACITOR VALUES ARE IN µf
RESISTOR VALUES ARE IN OHMS
TOLERANCES ON
FRACTIONS ± DECIMALS ± ANGLES ±
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MATERIAL

MIT
INSTRUMENTATION LAB
CAMBRIDGE, MASS.

DRAWN	ADM James A. DelVecchio	29 Jun 68
CHECKED	[illegible]	12 Aug 68
APPROVED	REB Eldon C Hall	12 Aug 68
APPROVED	D.G. Hoag	12 aug 68
APPROVED CCB	S Baca	8/20/68
APPROVED MSC		8/1?/68
		DATE

MANNED SPACECRAFT CENTER
HOUSTON, TEXAS

INDICATOR
STATUS/CAUTION
SPECIFICATION CONTROL DRAWING

CODE IDENT NO.	SIZE	DRAWING NO.
80230	C	1005025

SCALE **NONE** SHEET **1** OF **3**

MODERN BLUE PRINT CO., INC. 02840

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REQUIREMENTS: (CONTINUED)

3. DESIGN:

A. OPERATING LIFE: 4000 HOURS MINIMUM AT POWER DISSIPATION PER NOTE 2.B.(1) AT 70° ± 10° F AMBIENT TEMPERATURE.

B. MATERIAL: ALL MATERIAL SHALL BE CAPABLE OF MEETING THE REQUIREMENTS OF THIS DRAWING AND EXHIBIT NO TOXICITY.

(1) PLASTIC FACE PLATE: ACRYLIC SHEET PLASTIC PER MIL-P-5425, FINISH A.

(2) CASE: ALUMINUM ALLOY.

C. FINISHES:

(1) ALUMINUM CASE: CHROMATE TREAT PER MIL-C-5541.

(2) PAINT SURFACES MARKED "A" AND TOP BORDER AREA (EXCLUDING THE SURFACE OF PANELS) GREY EPOXY ENAMEL PER ND 1002279 USING 1008809-1.

(3) FINAL COAT SURFACES MARKED "A" AND TOP BORDER AREA (EXCLUDING THE SURFACE OF PANELS) PER ND 1002277 USING 1012543-003.

D. TERMINALS AND INSULATORS: SHALL MEET THE REQUIREMENTS OF ND 1002136.

E. PANEL COLORS: (REF. NOTE 2.A.(4)) YELLOW COLOR TO BE AVIATION YELLOW AND WHITE COLOR TO BE AVIATION WHITE PER MIL-C-25050. COLOR COORDINATES SHALL REMAIN WITHIN THE LIMITS OF THE SPECIFIED AVIATION COLORS WHEN ENERGIZED FROM 5.0 THROUGH 3.0 VDC. THE X - COORDINATE FOR THE AVIATION WHITE DISPLAYS SHALL NOT EXCEED X = .560 WHEN ENERGIZED FROM 5.0 THROUGH 2.0 VDC.

COLOR, UNENERGIZED: LEGEND BACKGROUND TO BE LOW REFLECTIVE, DIFFUSE, NEUTRAL GREY WHEN NOT ENERGIZED. CONTRAST BETWEEN BLACK CHARACTERS AND UNLIGHTED BACKGROUND SHALL BE BETWEEN 2 TO 4 UNITS IN ACCORDANCE WITH MIL-P-7788. EXCEPT CONTRAST, $C = \frac{B_1 - B_2}{B_2}$

SURFACE GLOSS: THE DISPLAY AREAS SHALL HAVE A SURFACE GLOSS OF 5 UNITS OR LESS AS MEASURED PER MIL-P- 7788 WHEN IN THE UNENERGIZED CONDITION.

F. LAMPS: EACH OF THE 14 LEGEND AREAS SHALL BE ILLUMINATED BY 3 LAMPS PARALLEL WIRED AS SHOWN IN THE SCHEMATIC. YELLOW DISPLAY LAMPS TO BE MS 24367-680. WHITE DISPLAY LAMPS TO BE MS 24367-713.

G. WHEN MOUNTED ON A SUITABLE FIXTURE, THE UNIT SHALL BE CAPABLE OF WITHSTANDING A PRESSURE OF 15 PSIA OF HELIUM APPLIED AGAINST THE 32 AREA, WHEN THE EXTERNAL PRESSURE IS REDUCED TO 1" HG EQUIVALENT PRESSURE: WHEN SUBJECTED TO THIS DIFFERENTIAL PRESSURE, THE UNIT SHALL EXHIBIT A LEAK RATE OF NO MORE THAN 1×10^{-5} ATM CC/SEC OVER THE TEMPERATURE RANGE OF -10°C TO +82°C.

4. NOTE: THIS UNIT IS TO BE MOUNTED ON A HEAT SINK WHEN LEGENDS ARE ENERGIZED FOR PERIODS LONGER THAN TWO (2) MINUTES. A SATISFACTORY HEAT SINK CAN BE AN ALUMINUM OR BRASS PLATE 10" X 10" X 1/8" WITH A CENTERED RECTANGULAR OPENING FOR MOUNTING THE UNIT.

5. CONSTRUCTION:

A. BULBS TO BE USED SHALL MEET THE REQUIREMENTS OF MS 24367, AND SHALL BE SCREENED FOR THE FOLLOWING DEFECTS:

(1) SURFACE CRACKS.

(2) SCRATCHES AND TOOL MARKS.

(3) RAW UNPOLISHED GLASS EDGES IN FINAL SEAL AREA.

BULBS WHICH EXHIBIT THESE DEFECTS SHALL NOT BE USED IN DISPLAYS SUPPLIED TO THIS SPECIFICATION.

B. BULBS SHALL BE CONFORMALLY COATED WITH A SOFT SETTING PLASTIC BEFORE EMBEDDMENT TO RELIEVE POTTING STRESS ON BULBS. THE PLASTIC AND/OR SOLVENTS SHALL BE AS NEGOTIATED PER ND 1015404, CLASS NO. 2.

6. VENDOR SUPPLIED DOCUMENTATION:

THE VENDOR SHALL RECORD AND SUPPLY CERTIFICATES OF COMPLIANCE DATA FOR THE FOLLOWING READINGS ON EACH DISPLAY

A. BRIGHTNESS PER PARAGRAPH 2.A.(5)

B. POWER DISSIPATION PER PARAGRAPH 2.B(1).

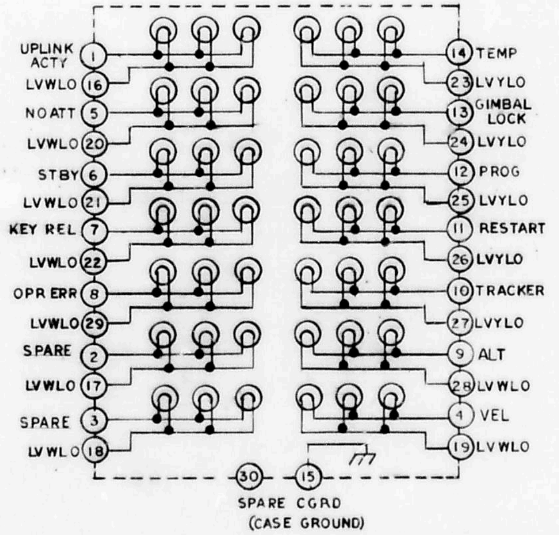
C. DIELECTRIC WITHSTANDING VOLTAGE PER PARAGRAPH 2.B(2).

D. INSULATION RESISTANCE TEST PER PARAGRAPH 2.B(3).

E. ISOLATION PER PARAGRAPH 2.A(6).

F. PANEL COLOR AND MARKING PER PARAGRAPH 2.A(4).

7. BULB BORN-IN: BULBS SHALL BE BURNT-IN FOR A MINIMUM OF 300 HOURS, AT 5.0 VDC.



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DRAWN	29 Jul 68	**INDICATOR** **STATUS/CAUTION** **SPECIFICATION CONTROL DRAWING**	
CHECKED	12 Aug 68		
APPROVED Eldon C Hall	12 Aug 68		
APPROVED D. G. Hoag	12 Aug 68		
APPROVED ~~MIT~~ CCB	8/20/68	CODE IDENT NO. 80230	SIZE C
APPROVED MSC	DATE	SCALE NONE	DRAWING NO. 1005025 SHEET 2 OF 3

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REVISIONS TDRR 36704

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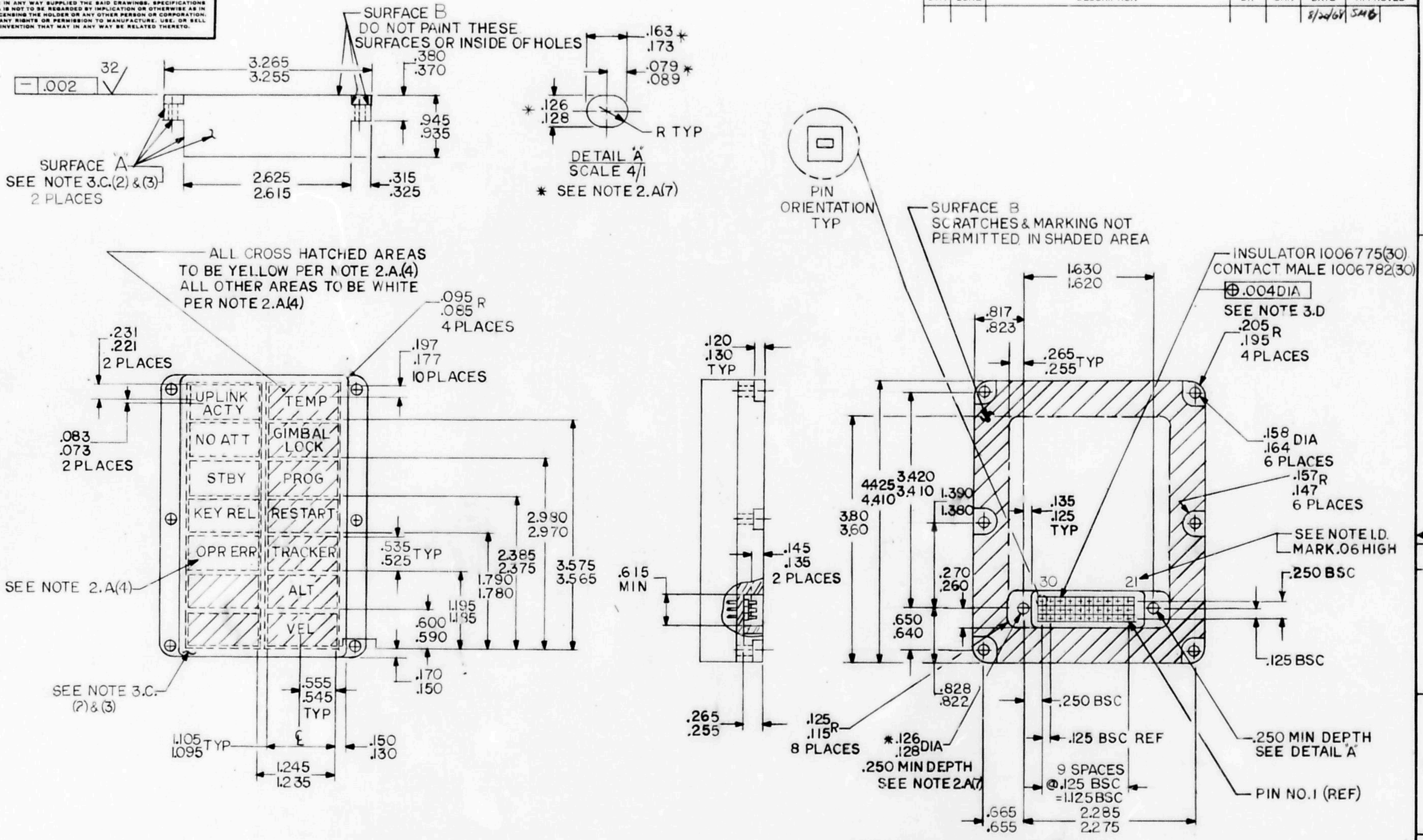
					8/22/68	SMB



.002
32
3.265
3.255
SURFACE B
DO NOT PAINT THESE SURFACES OR INSIDE OF HOLES
.380
.370
.945
.935
SURFACE "A"
SEE NOTE 3.C.(2) & (3)
2 PLACES
2.625
2.615
.315
.325
.163 *
.173
.079 *
.089
* .126
.128
R TYP
DETAIL "A"
SCALE 4/1
* SEE NOTE 2.A(7)
PIN ORIENTATION TYP
SURFACE B
SCRATCHES & MARKING NOT PERMITTED IN SHADED AREA
ALL CROSS HATCHED AREAS TO BE YELLOW PER NOTE 2.A.(4)
ALL OTHER AREAS TO BE WHITE PER NOTE 2.A.(4)
.095
.085 R
4 PLACES
.231
.221
2 PLACES
.197
.177
10 PLACES
UPLINK ACTY
TEMP
.083
.073
2 PLACES
NO ATT
GIMBAL LOCK
STBY
PROG
KEY REL
RESTART
OPR ERR
TRACKER
ALT
VEL
SEE NOTE 2.A(4)
.535
.525 TYP
2.980
2.970
2.385
2.375
3.575
3.565
1.790
1.780
1.195
1.185
.600
.590
.170
.150
SEE NOTE 3.C. (2) & 3
.555
.545
TYP
1.105
1.095 TYP
℄
.150
.130
1.245
1.235
.120
.130
TYP
.615
MIN
.145
.135
2 PLACES
.265
.255
.125
.115 R
8 PLACES
* .126
.128 DIA
.250 MIN DEPTH
SEE NOTE 2.A(7)
1.630
1.620
INSULATOR 1006775(30)
CONTACT MALE 1006782(30)
⊕ .004 DIA
SEE NOTE 3.D
.817
.823
.205
.195 R
4 PLACES
.265
.255 TYP
.158
.164 DIA
6 PLACES
.157
.147 R
6 PLACES
4.425
4.410
3.420
3.410
1.390
1.380
3.80
3.60
.135
.125
TYP
SEE NOTE 1.D.
MARK .06 HIGH
.270
.260
30
21
.250 BSC
.650
.640
.125 BSC
.828
.822
.250 BSC
.125 BSC REF
.250 MIN DEPTH
SEE DETAIL "A"
9 SPACES
@ .125 BSC
= 1.125 BSC
PIN NO. 1 (REF)
.665
.655
2.285
2.275

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DRAWN	James A. McCracken	24 Jul 68
CHECKED		12 Aug 68
APPROVED	Ellen C. Hall	12 Aug 68
APPROVED	D. G. Hoag	12 Aug 68
APPROVED ~~NEXT~~ CCB	S. D.	8/22/68
APPROVED MSC		9/12/68
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SCALE 1/1 SHEET 3 OF 3

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